

Self-diffusion of water and poly(amidoamine) dendrimers in dilute aqueous solution†

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Medium generation PAMAM dendrimers are extensively researched as drug delivery vehicles therefore detailed knowledge of their physico-chemical properties in solution is vital. We have selected ethylenediamine core, generation five poly(amidoamine) (PAMAM_E5) dendrimers (amine terminated PAMAM_E5.NH₂ and its succinamic acid derivative PAMAM_E5.SAH, respectively) as model compounds to study their dynamic behavior in water as a function of pH and concentration. Diffusion coefficients of water and the dendrimers were determined in deuterium oxide using pulse-field gradient stimulated echo (PGSE) NMR. The diffusion rate for PAMAM_E5.NH₂ increased monotonously with increasing pH values while for PAMAM_E5.SAH a maximum was found at an isoelectric value of pH = 5.7. The apparent diffusion coefficients of dendrimers decreased linearly with increasing concentration measured at their self-pH (pH = 9.4 for PAMAM_E5.NH₂ and pH = 5.7 for PAMAM_E5.SAH) in the absence of added salts. The observations could be explained by considering hard sphere interactions between strongly hydrated dendrimer molecules. The average hydrodynamic radii of dendrimers were determined by extrapolating the measured diffusion coefficients to zero dendrimer concentration and applying the Stokes–Einstein equation. The calculated values were $R_H = 3.05 \pm 0.04$ nm for PAMAM_E5.NH₂ and $R_H = 3.37 \pm 0.08$ nm for PAMAM_E5.SAH respectively. Measured diffusion coefficients of water (D₂O) also decreased linearly with increasing dendrimer concentration. From the concentration dependence, the average number of water molecules that form one dynamic unit with one macromolecule could be calculated on the basis of three different obstruction models. It has been concluded that hydrated PAMAM dendrimers in aqueous solutions behave as soft colloids against solvent molecules but as hard-sphere colloids against each other. Their equivalent hard-sphere radii were found to be equal to the measured hydrodynamic radii.

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Introduction

For *in vitro* and *in vivo* applications and for theoretical considerations it is important to understand both motion and diffusion of organic macromolecules in solution.¹ Dendrimers are nearly monodisperse, symmetrically branched macromolecules with well-defined structures. Their medium and higher generations have close to spherical geometry in solution and are actively researched as carriers of imaging and therapeutic agents.^{2–7} Polyionic dendrimers, *e.g.* poly(amidoamine)s (PAMAMs) and poly(propyleneimine) (PPI) dendrimers are often a subject of experimental and theoretical investigations because of their symmetry and relative simplicity as colloidal

particles.^{1,3–5,8–22} To correctly interpret dendrimers' capability to transport and release physiologically active small molecules we must understand their dynamic interactions with water.^{8,9} Depending on their family and generation, dendrimer molecules in solution may behave as “small” branching molecules, “soft” colloids (draining networks, *etc.*), or “hard” organic nanoparticles. Low generation dendrimer molecules (LGD) are relatively small molecules or low molecular mass polymers (oligomers) with fully penetrable networks and a few reactive sites that are all equally accessible for solvent molecules (the termini are all on the “surface” of the dendrimer molecules).² High generation dendrimer molecules (HGD) are spherical organic nanoparticles with high structural symmetry and high relative density. In these dense organic nanoparticles, the kinetic accessibility of groups on the surface is much greater than that of groups in the interior space. Medium generation dendrimer molecules (MGD) are between LGD and HGD and display characteristics of both groups. These dendrimers are described to behave as hybrids between polymer chains and compact colloidal particles based on extensive theoretical calculations and experimental studies.^{8,10,23} It is well known that

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the charge, inside structure and size of polyionic dendrimers change as a function of pH or concentration. All these factors affect the diffusivity of these macromolecules; therefore studying their diffusion behavior in solution can be a unique source of information beyond the almost exclusively used small angle scattering methods.²³ Pulse-field gradient stimulated echo (PGSE) NMR spectroscopy in combination with other NMR methods for solution structure and dynamics is an excellent method of choice to obtain data for solution behaviors of macromolecules in general and for those of polyionic dendrimer model systems as well.^{24–27}

Soon after the first synthesis of arborol dendrimers (alkane core) the influence of pH on their size was observed.¹¹ It was reported that the size of third generation amine-terminated arborol dendrimers (determined from diffusion coefficients by the Einstein–Stokes equation) increased with decreasing pH.²⁸ This “swelling” was explained by the Coulomb repulsion between the protonated primary amine termini. Arborol dendrimers with carboxylic acid terminal groups behaved the opposite way, as deprotonation of terminal carboxylic groups caused an increase in size with decreasing pH. A small decrease was observed in basic solution with increasing pH but no explanation was provided.^{11,28} In the case of PPI (poly-propylene-imine) dendrimers with carboxylic acid termini a size minimum appeared as a function of pH.²⁹ The explanation was that these PPI dendrimers contained not only carboxyl termini, but proton acceptor tertiary amino groups in the bifurcation points. At pH = 6, the formally neutral dendrimer exhibited a minimum size, while both at pH = 2 (positively charged) and pH = 11 (negatively charged) larger hydrodynamic radii were observed. Increasing hydration with the increase of charge was suggested to explain the larger size of dendrimers at low and at high pH, although the electrostatic repulsion of charged atoms was also considered as an alternative.²⁹

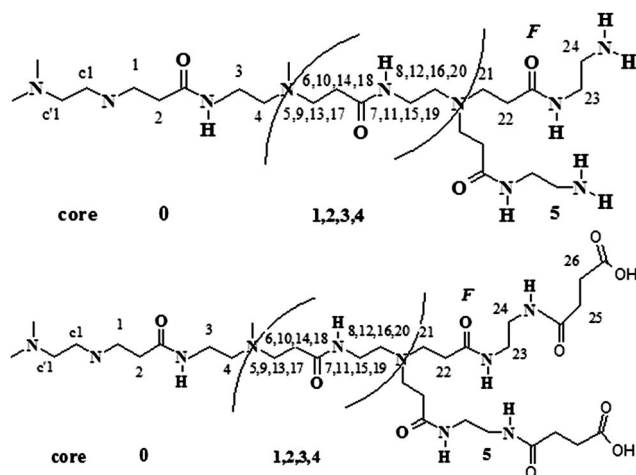
Goddard and his co-workers reported extensive molecular dynamic calculations predicting variation in the size of EDA (ethylenediamine) core PAMAM_Ex.NH₂ dendrimers as a function of pH.^{14,15} They have calculated that the size of amine terminated PAMAM dendrimers increases with decreasing pH. Their results were interpreted as due to a significant back-folding of “surface” amine groups into the interior of the macromolecules, even at low pH when both the tertiary and primary nitrogens are protonated. They found that the electrostatic repulsion between charged terminal –NH₃⁺ groups caused only a small increase in size around neutral pH on the model used. At low pH, when the tertiary nitrogens were fully protonated, a larger increase in size was calculated. Penetration of water molecules into the interior of dendrimers was found to be an important factor in swelling of the molecules.¹⁵ Most recently refined atomistic MD simulations showed that the radius of gyration of fourth generation PAMAM_E4.NH₂ dendrimer molecules was essentially pH independent in accordance with earlier experiments but the molecules underwent pH responsive conformational changes.^{30,31}

Lin *et al.* performed molecular dynamic simulations on properties of water molecules inside the PAMAM dendrimers. The results were interpreted in terms of three types of solvent

molecules such as “buried”, “surface” and “bulk” ones, characterized by their number, dynamic behavior and by their spatial position in or around the dendrimer molecules.¹³

The change of the apparent diffusion coefficient as a function of total concentration also provided useful information about the size and the nature of interactions between dendrimer molecules although the interpretation of data was not straightforward.^{32,33} In dilute methanol solution (volume fraction less than 0.2) the diffusion rate of PPI dendrimers from 1 to 5 generations decreased linearly with the increase of the total concentration. An effective hard sphere radius could be determined for PPI dendrimers, which was larger than the radius of gyration but smaller than the hydrodynamic radius. From these results Rietveld and Bedeaux concluded that *in dilute methanol* solution no “soft colloid” interaction exists between dendrimer molecules.³² The scaling behavior and hydrodynamic radii of G0–G3 (E0–E3) PAMAM.NH₂ dendrimers were determined by PGSE NMR in water from the concentration dependence.³⁴

In order to gain deeper insight into the interaction, size variation and hydration of PAMAM dendrimers as a function of pH and total concentration in aqueous solution, we performed extended PGSE NMR studies to determine the diffusion coefficients for both dendrimers and water molecules in the same solution. We have chosen two EDA core, generation five PAMAM dendrimers with characteristically different termini: an amine terminated compound (PAMAM_E5.NH₂), and a succinamic acid derivative (PAMAM_E5.SAH) (Scheme 1). Our goal was to study the application of colloid suspension theories for dilute aqueous solutions of hydrated PAMAM dendrimers of medium generation.²³ We expected a characteristic pH dependence of the diffusion coefficient of PAMAM_E5.SAH because between pH = 2–8 this dendrimer changes its net charge from positive to negative as a function of pH. To determine the hydration of dendrimer molecules experimentally we implemented the method described for surfactants and colloidal systems.^{35,36} To



Scheme 1 Structural formula of an ideal dendron of PAMAM_E5.NH₂ (upper) and PAMAM_E5.SAH dendrimers. The numbers indicate both the carbons and their respective protons for NMR peak assignment. The boldface numbers under the formula are generation numbers.

our knowledge this is the first report of experimental hydration data for PAMAM dendrimers; only a rough estimation was published before.³⁷

Experimental

Materials

Ethylenediamine (EDA) core PAMAM dendrimers of generation 5 (PAMAM_E5.NH₂, 1205_06-E5.0-LD for short E5.NH₂) were purchased from Dendritech (Midland, MI, USA) in methanol solution containing 14.17% E5.NH₂. The methanol was removed using a rotavap, and the resulting solid was dissolved in water, dialyzed against distilled water using a MWCT = 10 000 membrane (Pierce Scientific). This E5.NH₂ dendrimer material contained 89.64% E5.NH₂ as the major fraction, 8.11% (E5.NH₂)₂ dimers, and 2.25% of E4.NH₂ dendrimers as the trailing generation. Its average molecular mass was equal to M_w , (SEC) = 30 720, M_w/M_n = 1.05. The PAMAM_E5.SAH (for short E5.SAH) was synthesized and characterized as it is written in our previous papers.^{38–40} The measured data of molecular weight and polydispersity index: M_w (SEC) = 40 710, M_w/M_n = 1.045. KCl, NaOH, KOH, HCl and deuterium-oxide were purchased from Sigma-Aldrich and used without further treatments.

Sample preparation

For diffusion studies, solutions of different total concentrations of dendrimers were prepared between c = 0.2–1.1 mmol kg^{−1} for E5.SAH and c = 0.4–1.5 mmol kg^{−1} for E5.NH₂ in D₂O. The weighed amount of lyophilized dendrimers was dissolved in D₂O, and diluted with the weighed amount of D₂O. No further concentration determination was performed. Relative concentrations were checked by means of ¹H NMR integrals. For studying the pH dependence the lyophilized E5.NH₂ dendrimer was dissolved in an appropriate amount of DCl solution. E5.SAH was dissolved in D₂O which resulted in slightly acidic solution (pD ≅ 5). The pH was set in both samples by using concentrated NaOH (in D₂O) and concentrated HCl solutions to avoid substantial dilution. pD values were read and the pH was calculated as pH = pD + 0.415.^{41,42}

Instruments

The pH was measured using a Metrohm 877 pHlab pH meter using a Metrohm 6.0234 combined glass electrode. The pH potentiometric titration was performed with Metrohm 736 GP Titrino automatic titration systems in a volume of 4 cm³. 0.2 mol dm^{−3} KOH, standardized on KH-phthalate, was used for titration of acidified dendrimer solutions. The ionic medium was constant, I = 0.2 M KCl during the titration. The carbon dioxide was kept out by bubbling argon through the solution. The H⁺-ion concentration was determined from the measured pH by Irving's method.⁴²

1D and 2D, ¹H, and ¹³C NMR experiments were performed usually at 300 K with Bruker DRX 400 and 360 MHz instruments using standard pulse programs provided by the manufacturer (see ESI†) normal 5 mm direct QNP and inverse BB (z gradient)

probe heads. The temperature was calibrated by the ¹H NMR chemical shift of methanol.⁴³ This temperature could be kept constant by cooling air flow of the lab temperature and electric heating controlled by an Eurotherm unit. We post-processed the spectra with Bruker WinNMR® software.

For diffusion measurements the pulse field gradient stimulated spin echo sequence (PGSE) was used with longitudinal eddy-current delay (LED) and bipolar gradient pulses (BIPLED) by the standard program provided with the spectrometer.⁴⁴ The diffusion of dendrimer was measured on peaks of CH₂ protons, while the water diffusion (D₂O) on HOD peak. Deuterium NMR would have been an option as well, but ¹H NMR is technically easier and provides us the same information. In diffusion measurements, we used a constant diffusion time (Δ), typically 10–15 ms, and gradient pulse length (δ), 4 ms, for water, while 50–60 ms and 8–10 ms for dendrimers. In general, 3–5 parallel measurements were performed on one sample with different combinations of the above parameters. The pulsed gradient strength (G) increased with 16 square distant steps from 0 to approximately 50 Gauss cm^{−1}. We accepted the calculated diffusion coefficients when the values differed less than 1% in the case of water and less than 3% in the case of dendrimers.

Data evaluation

Potentiometric titration data were evaluated with the Hyperquad program package.⁴⁵ In order to obtain equilibrium data both solution complex formation models and surface complex formation models were considered.⁴⁶ We performed titrations at different total concentrations; primary titration curves and the detailed results are shown in the ESI.†

The diffusion coefficient is determined from the attenuation of echo signals by increasing the pulsed gradient strength (G) as shown in the ESI,† according to eqn (1).⁴⁴

$$I = I_0 \exp(-\gamma^2 G^2 \delta^2 (\Delta - \delta/3) D_{\text{obs}}) \quad (1)$$

where I is the integrated intensity of the NMR peaks and γ is the gyromagnetic ratio. The non-linear least squares method was used to determine the diffusion coefficient, D_{obs} in individual units. The calculation of the real numeric values of diffusion coefficients (D calibration) was performed using the known diffusion coefficient of D₂O at the given temperature.⁴⁷ From the diffusion coefficients the hydrodynamic radii (R_H) of dendrimers can be calculated by the Stokes–Einstein equation:

$$R_H = \frac{k_B T}{6\pi \eta D} \quad (2)$$

assuming that the dendrimers are spherical and the medium is a continuum. In eqn (2) symbols have their usual meaning.

From the diffusion coefficient of water in the presence of dendrimer, the number of water molecules moving with dendrimers can be determined according to the method introduced by Nilsson and Jönsson and their co-workers.^{35,36} They consider two extreme situations. The first case is when the colloidal particles obstruct the diffusion of water but otherwise do not influence either their local concentration profile or local diffusion coefficient. Based on the extension of the cell model of

liquid state to solution of non-charged colloidal particles the following equation is obtained:

$$D_w^{\text{obs}} = D_w^0 \frac{1}{1 + \phi/2}, \quad (3)$$

where D_w^0 and D_w^{obs} are the measured diffusion coefficients of water in the absence and in the presence of dendrimer of volume fraction ϕ . The second case is when the water molecules interact with colloidal particles. In the simplest model we distinguish free and bound water molecules and the measured diffusion coefficient of water is expressed as:

$$D_w^{\text{obs}} = p_{\text{free}} D_w^0 + p_{\text{bound}} D_d^0 \quad (4)$$

where p is the molar fraction of the “free” and “bound” water while D_d^0 is the diffusion coefficient of the colloidal particle (dendrimer). Eqn (3) and (4) were applied in combination introducing an apparent concentration dependence in the molar fraction of free water.³⁶

$$D_w^{\text{obs}} = p_{\text{free}} \frac{D_w^0}{1 + \phi/2} + p_{\text{bound}} D_d^0 \quad (5)$$

Results and discussion

Potentiometric data

Protonation constants and the average number of the functional groups of dendrimers were determined from the titration curves. For E5.NH₂ a simple solution equilibrium model sufficiently describes the titration curves. In this model the primary amines and the tertiary nitrogen atoms are considered as independent protonation sites. The best fit of data results in an average 119 tertiary nitrogen atoms and 122.6 primary nitrogen atoms (formula values are 126 and 128 respectively). The two pK values are 5.72(2) and 8.90(2) in 0.2 mol dm⁻³ KCl, similar to those published by Cakara *et al.*^{48,49} We performed the same fitting procedures for E5.SAH. The pK values are 4.60(1) and 6.69(1) corresponding to -COOH 117 and tertiary nitrogen 108. The pK of carboxylate termini are somewhat higher, while the pK of tertiary nitrogens are considerably lower than those published for carboxylated PPI dendrimers.⁵⁰ Although the literature describes other methods to analyze the titration curves (see ESI†),^{48,50,51} according to the sensitivity of our titration the donor groups seem to bind protons quite independently in both dendrimers and this simple approach is satisfactory for the qualitative interpretation of diffusion properties.

High resolution NMR results

¹H NMR spectra of dendrimers are usually poor in signals as a consequence, the fine structure of the ¹H NMR spectra caused by scalar couplings is not always observed as shown in Fig. 1. Thus assignment and structural analysis requires a lot of care and good technical background. 2D COSY spectra were used to assign scalar coupling protons, being present on neighboring carbons.^{52,53} ¹³C NMR spectra through HSQC and HMBC were used for complete assignment. The reason for the poor

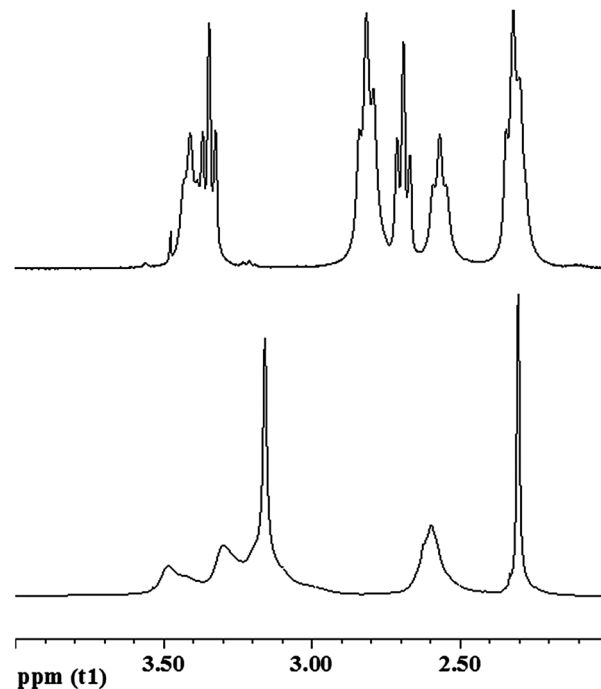


Fig. 1 ¹H NMR spectra of PAMAM_E5.NH₂ (upper) and PAMAM_E5.SAH solutions in D₂O at pH = 10.8 and pH = 5.4 respectively. Spectra were recorded at their “self-pH”, where both dendrimers are formally neutral.

resolution is that the protons in identical structural positions but in different shells (*e.g.* 8, 12, 16 and 20) are not perfectly the same; slight differences in chemical shifts may occur. The relaxation broadening of the peaks is often large enough to mask completely the fine structure of the spectrum.

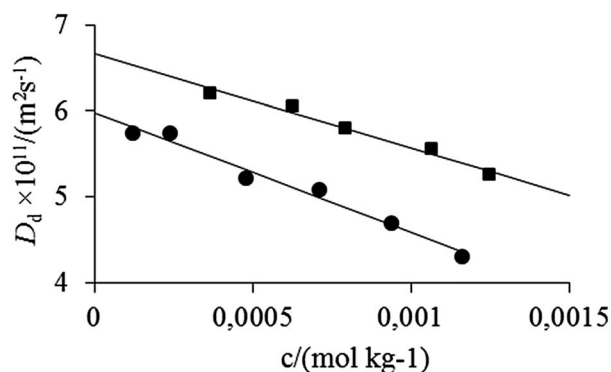
In Table 1 the assignment of protons is listed while the step-by-step process is described in the ESI.† There is an eye-catching difference between the two spectra: E5.NH₂ has narrower peaks; even some scalar couplings can be estimated from the 1D spectrum while the spectrum of E5.SAH has no fine structure at all. The probable reason is that the formally neutral E5.SAH at its point of zero charge (PZC) consists of positively (-NH⁺(*tert*)) and negatively charged groups (COO⁻) in almost equal amounts. The internal charge compensation may result in a compact macromolecule with only one, long global rotation correlation time yielding a short transverse relaxation time and broad ¹H NMR peaks. E5.NH₂ at its self-pH is electrically neutral. In this case the much weaker internal attractive forces permit faster local motion and consequently longer local transverse relaxation times (narrower lines) for certain CH₂ groups. This explanation is supported by the values of ³J H-H scalar coupling constants which are between 6 and 7 Hz, typical values for freely rotating CH₂-CH₂ units.⁵⁴

Diffusion of dendrimers

In Fig. 2 the measured diffusion coefficients of the two PAMAM dendrimers as a function of concentration are shown. The hydrodynamic radius R_H for E5.NH₂, calculated from the values at zero concentration, is 3.05 nm in agreement with literature values but much larger than the radius of 2.15 nm calculated

Table 1 Assignment of ^1H NMR peaks of dendrimers

PAMAM_E5.NH ₂			PAMAM_E5.SAH		
Shift/ppm	Numbers in Scheme 1	Note	Shift/ppm	Numbers in Scheme 1	Note
2.48	2,6,10,14,18,22	$J = 6.8$ Hz (br)	2.42	25–26	Singlet
2.67	4,8,12,16,20	$J \sim 6.4$ Hz (br)	2.72	2,6,10,14,18,22	Broad
2.77	24	$J = 6.2$ Hz	3.28	23–24	Singlet
2.87	1,5,9,13,17,21	$J = 6.8$ Hz (br)	~ 3.28	3,7,11,15,19	Broad
3.29	23	$J = 6.2$ Hz	3.42	1,5,9,13,17,21	Broad
3.34	3,7,11,15,19	$J \sim 6.4$ Hz (br)	3.6	4,8,12,16,20	Broad

**Fig. 2** The measured diffusion coefficients of dendrimers as a function of the molar concentration of E5.SAH (lower curve) and E5.NH₂ (upper curve) dendrimers. $T = 300$ K.

from TEM experiments, proving that PAMAM dendrimers are strongly hydrated.^{55,56} $R_H = 3.37$ nm is given for E5.SAH. The reason for the decrease of the diffusion rate with increasing concentration is the interactions between the dendrimer molecules even in these dilute solutions. The square root of the mean squared displacement of dendrimers is about $2\text{--}3\text{ }\mu\text{m}$ ($\langle(z - z_0)^2\rangle = (2Dt)$) during the typical diffusion time ($50\text{--}60$ ms).⁵⁷ If we calculate the volume fractions from the hydrodynamic radius (*see below*) and employ the cell model we can see that every dendrimer molecule sits in the middle of a water sphere of $6\text{--}10$ nm radius ($0.0015\text{--}0.0002\text{ mol dm}^{-3}$).³⁵ Therefore during the self-diffusion it crosses energy barriers of $150\text{--}200$ other ones.

Consequently, we determined *long-time self-diffusion coefficients* (dynamics of one dendrimer molecule under the influence of surrounding dendrimer molecules). The hard sphere colloid model of diffusion results in a decrease of the apparent diffusion coefficient with increasing concentration by the following equation:^{32,57}

$$\frac{D_d}{D_d^0} = 1 - 2.1\phi \quad (6)$$

where D_d is the apparent self-diffusion coefficient D_d^0 is the diffusion coefficient extrapolated to zero concentration while ϕ is the volume fraction of dendrimers.

There are three ways to calculate the volume fraction for E5.NH₂: from the hydrodynamic radius (ϕ_H), from the hard

sphere radius determined by SANS⁵⁸ (ϕ_{HS}) and from the volumes of constituent atoms, known as $18\text{ }\text{\AA}^3/\text{heavy atom}$ rule (ϕ_M).⁵⁹ In Fig. 3 and 4 the relative diffusion coefficients are plotted against volume fractions and the calculated slopes are given in Table 2. For E5.SAH the hard sphere radius from SANS is not available. It is seen that the slopes of D_d/D_d^0 vs. ϕ_H curves show good agreement with the most recent hard sphere model calculations shown in eqn (6).^{57,60} In this model the interaction energy of dendrimer molecules is zero ($V = 0$) when the distance between

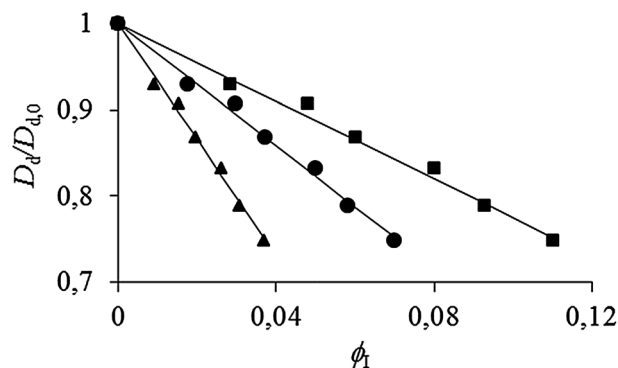
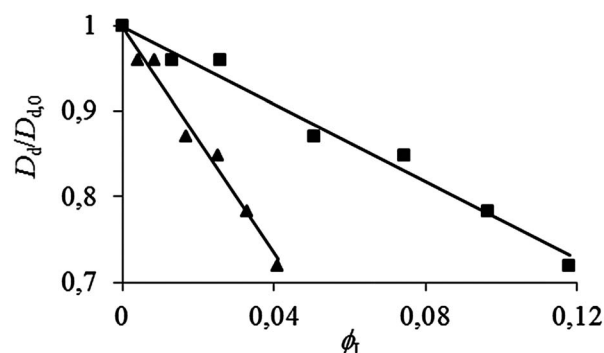
**Fig. 3** Dependence of the relative diffusion rate of E5.NH₂ as a function of volume ratio ϕ . (■) is the hydrodynamic volume fraction ($I = H$), (●) is the hard sphere volume fraction ($I = HS$), and (▲) is the volume fraction from specific volumes of dendrimers ($I = M$), in the same experiment.**Fig. 4** Dependence of the relative diffusion rate of the PAMAM_E5.SAH dendrimer as a function of volume ratio ϕ . (■) is the hydrodynamic volume fraction ($I = H$) and (▲) is the volume fraction from specific volumes of dendrimers ($I = M$), in the same experiment.

Table 2 The slopes of the diffusion coefficients dependence on the different volume fractions ϕ_i

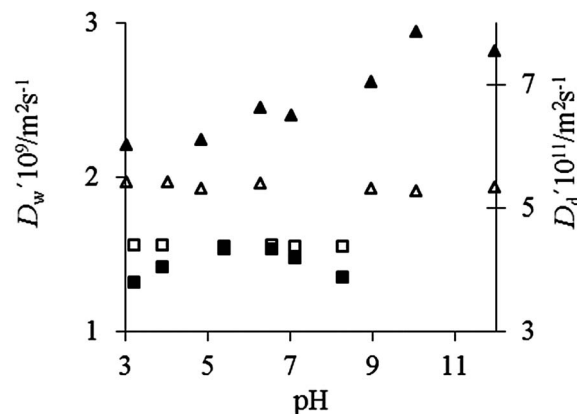
Type of volume fraction	Slope E5.NH ₂	Slope E5.SAH
Hydrodynamic	-2.1 ± 0.2	-2.3 ± 0.3
Hard sphere ⁵⁸	-3.5 ± 0.4	No R_{HS} data
Specific volume	-6.7 ± 0.6	-6.6 ± 0.9

the centers is $r > 2R$ where R is the radius. When the distance is smaller, $r \leq 2R$, the energy of repulsive interaction is $V = \infty$.

The reasons for the decrease of relative diffusion rate with increasing concentration are the hydrodynamic interactions resulting in an increase of the viscosity of the suspension, direct interaction and the Brownian force.⁵⁷ Fig. 4 and Table 2 show that E5.SAH behaves similarly. Our results confirm those ones described for diffusion of the fifth generation poly(propylene imine) dendrimers in methanol.³² However in that case the “effective hard sphere radius” was between the hydrodynamic radius and the radius of gyration. In our case both hydrodynamic radii are the “effective hard sphere radius”. The possible explanation is that PAMAM molecules are better solvated in water than PPI in methanol.

When the “hard sphere radius” of E5.NH₂ (determined by SANS⁵⁸ in dilute aqueous solution) is used to calculate the volume fraction (ϕ_{HS}) the negative slope is larger, -3.5 for E5.NH₂ (Fig. 2 and Table 2). If this calculation was correct this steeper decrease can be explained by additional interactions between the dendrimer molecules. “Soft” or “ultrasoft” interactions were introduced to describe this behavior by Likos and Ballauf *et al.*^{8,10,23,61} They have suggested that PAMAM dendrimers exhibit an effective interaction potential that may be described by a Gaussian function.^{62,63} However, this type of interaction probably exists in more concentrated solution of less solvating medium, than that applied in the present work. This interaction was observed experimentally for PPI dendrimers in methanol and poly(allylcarbosilane) dendrimers in chloroform at higher concentrations.^{32,64}

If we used ϕ_M volume fractions (calculated from density values) the slopes would be -6.6 and -6.7 (Table 2). Such a steep decrease of diffusion rate could only be explained by self-association.^{65,66} Applying Lévy's approximation for the linear dependence of D_d on the molar concentration (shown in Fig. 2) the slope is equal to $-2K(D_d^0 - D_{dimer})$, where K is the equilibrium constant and D_{dimer} is the diffusion constant of non-covalent dimeric association while D_d^0 is the intercept. The D_{dimer} may be estimated as $D_d/1.38$.⁶⁷ From the concentration dependence shown in Fig. 2 K can be numerically estimated as 120 (the slope is 1.1×10^{-8}) for E5.NH₂ and 160 (slope is 1.4×10^{-8}) for E5.SAH. These “dimerization” constants would mean that under our experimental conditions 20–30% of dendrimers were in dimeric associate form, which is improbable. Our conclusion is that the hydrated generation 5 PAMAM dendrimers behave like hard sphere colloids in dilute aqueous solutions. The effective hard sphere radius is equal to the hydrodynamic radius. The probable reason is the strong hydration which competes efficiently with the expected soft or ultrasoft interactions.

**Fig. 5** The pH dependence of the diffusion coefficient of D₂O (D_w , left axis) and the dendrimers (D_d , right axis). $\square$ and $\blacksquare$ show the diffusion of water and E5.SAH in the same solution respectively ($c = 0.41 \text{ mmol kg}^{-1}$, $T = 293 \text{ K}$) while Δ and $\blacktriangle$ show the diffusion water and E5.NH₂ in the same solution ($c = 0.37 \text{ mmol kg}^{-1}$, $T = 300 \text{ K}$) respectively.

The diffusion rates of dendrimers are pH dependent but the diffusion rate of water is insensitive to pH (Fig. 5). In the case of E5.NH₂, the increase of D_d with increasing pH leads to saturation after pH = 9. Earlier MD calculations indicated that the deprotonation of tertiary nitrogens caused a larger decrease in size than the deprotonation of terminal $-\text{NH}_3^+$ groups.¹⁵ Later, refinements with different force fields resulted in smaller pH sensitivities.³¹ Although our diffusion coefficients may be related to the change of hydrodynamic radius with pH, we cannot calculate R_H here from D values due to the lack of viscosity data. (The adjustment of pH was managed by adding HCl (vapor of conc. HCl solution) and concentrated NaOD solution without the control of the salt concentration.) The pH dependence should be studied in ionic medium, but in that case the role of the electric double layer must be considered. The data show in Fig. 5 that the hydrodynamic size is pH sensitive and the deprotonation causes a decrease in the size. From these experiments we cannot distinguish the role of tertiary and primary nitrogen in changing the size. For E5.SAH there is a maximum in the observed diffusion constant at isoelectric pH (PZC) indicating the smallest size. This is similar to what Rietveld *et al.* found for PPI-carboxylate dendrimers in water measured at three pH values.²⁹ For E5.SAH, the measured diffusion coefficients have much less uncertainty than that for E5.NH₂, because the self-pH is the PZC; therefore a much less amount of electrolyte had to be added to regulate the pH. Since the pH-dependent structure and dynamics of PAMAM dendrimers are not available the detailed experiments are in progress in our laboratory.

Diffusion of water

In Fig. 6, the decrease in the relative diffusion rate of water ($D_w^{\text{obs}}/D_w^0(\phi = 0)$) with increasing concentration of dendrimers is shown. Since the diffusion rate of water is sensitive to temperature we used the values resulting from the linear extrapolation to zero dendrimer concentration for the calculation.⁴⁷ (This way,

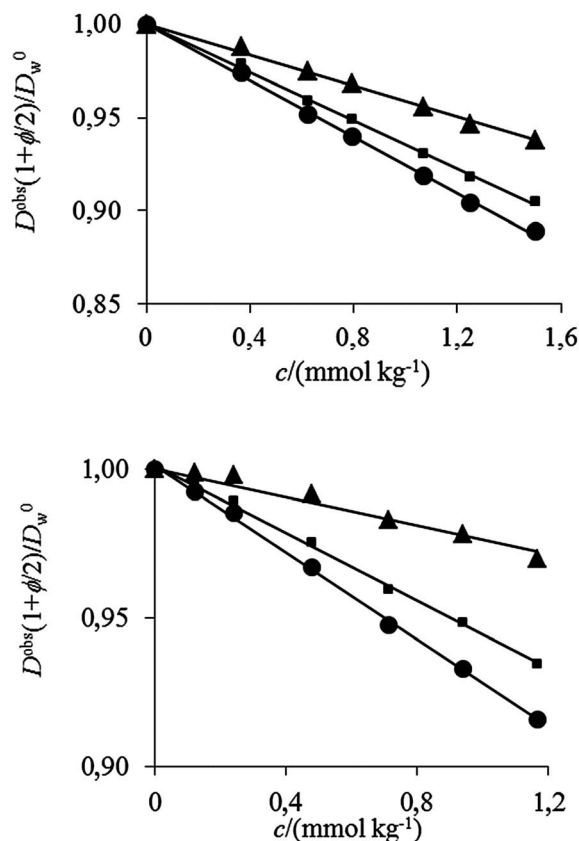


Fig. 6 Dependence of the corrected diffusion rate on the concentration of dendrimers E5.NH₂ (upper) and E5.SAH (lower) at 300 K: ● $\phi = 0$, ■ ϕ_M while ▲ ϕ_H .

we can eliminate the effect of small temperature differences between experiments, which might occur even at the same nominal temperature of the NMR probe head.) There are two possible explanations for the decrease for the observed diffusion rate of water. First is that the studied dendrimer molecules behave as hard colloids therefore they obstruct the diffusion of water molecules according to eqn (3). The other possible explanation is that we measure the average diffusion coefficient of D₂O; this value must be lower in the presence than in the absence of dendrimers according to eqn (4). The combination of the two effects may also be considered as eqn (5) shows. The obstruction effect can be predicted, provided we can apply the appropriate volume fraction of dendrimer.³⁵ From the NMR point of view there are only free and bound waters, that is $p_{\text{bound}} + p_{\text{free}} = 1$, therefore eqn (5) becomes eqn (7)

$$D_w^{\text{obs}} = p_{\text{bound}} \left(D_d^0 - \frac{D_w^0}{1 + \phi/2} \right) + \frac{D_w^0}{1 + \phi/2} \quad (7)$$

with

$$p_{\text{bound}} = \frac{n_d n_{\text{wd}}}{n_d n_{\text{wd}} + n_w} \quad \text{and} \quad c_d = \frac{m_d / M_d}{m_d + m_w} \quad (8)$$

where n_d is the number of moles of the dendrimer, n_w and n_{wd} are the number of moles of water in the bulk and in one mole dendrimer respectively; c_d is the dendrimer concentration in mol kg⁻¹ while m_d , M_d are the mass and molar mass of dendrimers and m_w is the mass of water.

Combination of eqn (7) and (8) leads to eqn (9) with the approximation that n_{wd} is small in the denominator

$$p_{\text{bound}} = \frac{c_d(m_d + m_w)n_{\text{wd}}}{c_d(m_d + m_w)n_{\text{wd}} + m_w/M_w} \approx \frac{c_d(m_d + m_w)n_{\text{wd}}}{m_w/M_w} \approx c_d n_{\text{wd}} M_w \quad (9)$$

then the comparison of eqn (7) and (9) with $D_d \ll D_w$ gives:

$$\begin{aligned} D_w^{\text{obs}} &\approx c_d n_{\text{wd}} M_w \left(D_d - \frac{D_w^0}{1 + \phi/2} \right) + \frac{D_w^0}{1 + \phi/2} \\ &\approx -c_d n_{\text{wd}} M_w \frac{D_w^0}{1 + \phi/2} + \frac{D_w^0}{1 + \phi/2} \end{aligned} \quad (10)$$

From eqn (10) the dependence of the measured diffusion coefficient *corrected with obstruction effect* shows a linear dependence on the molar concentration of dendrimers:

$$\frac{D_w^{\text{obs}}(1 + \phi/2)}{D_w^0} = 1 - n_{\text{wd}} M_w c_d \quad (11)$$

Earlier we used a simple formula, which appears frequently in the literature,⁶⁸ for calculating the molar fraction (number) of bounded water which suffered from inaccuracies: $p = (D_w^0 - D_w^{\text{obs}})/(D_w^0 - D_d)$.³⁷ The difference of 1% in the numerator of this equation may cause an error as large as ~500 in the number of water molecules. The basic idea behind the two approaches is the same.³⁵ It is also important to note that when we measure water diffusion using the ¹H signal of HOD, we are measuring a weighted average of signals of water protons and the *exchangeable* protons of dendrimers. The error introduced in this way is negligible; because the highest percentage of exchangeable protons of dendrimers is 0.5% of the measured HOD protons, therefore correction is not necessary.^{36,69}

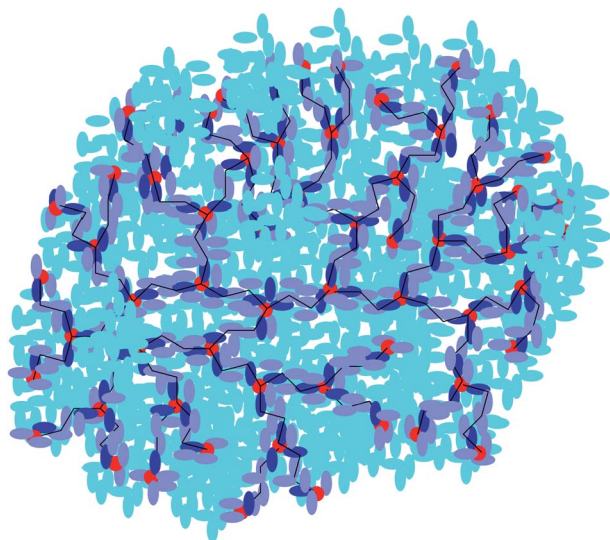
Fig. 6 shows the plots according to eqn (11) taking into account three models, *i.e.* three different ways of calculating the volume fractions of dendrimers. The numbers of water molecules, calculated from slopes, are collected in Table 3.

On the basis of Scheme 2 first we substitute the hydrodynamic volume fraction, ϕ_H into eqn (11). In this first and simplest model we use the hard sphere approximation and the diffusion obstruction (upper curves in Fig. 6).^{35,36} The number of water molecules (first row in Table 3) in this way is underestimated because we consider the total volume of hydrated dendrimer impenetrable for water, which is in contradiction with the accepted models of solvated polymers.^{2,4,58,68,70–72}

In the second model we assume that the whole dendrimer except the organic skeleton itself is penetrable for water

Table 3 The calculated molar volumes and the number of water molecules in one dendrimer molecule at 300 K

	E5.NH ₂ (pH = 10.8)		E5.SAH (pH = 5.42)	
Volume fraction	Volume (cm ³ mol ⁻¹)	No. of water molecules	Volume (cm ³ mol ⁻¹)	No. of water molecules
ϕ_H	70 816	2070 ± 70	98 400	1190 ± 30
ϕ_M	22 169	3077 ± 100	32 998	2760 ± 80
$\phi = 0$	0	3700 ± 100	0	3600 ± 100



Scheme 2 Illustration of models of hydration of the dendrimer molecule in aqueous solution. Ellipsoids stand for water molecules: dark blue ones bound to hydrophilic groups, lighter blue ones are obstructed by the dendrimer while the lightest ones are bulk water entrapped in dendrimer.

molecules. For this model we calculate φ_M from the specific volume or from the crystallographic volume (middle curves in Fig. 6, second row in Table 3).

The third model is a version of the draining network assumption. The hydrated dendrimers are imagined to behave as “water droplets in water”. One incoming water molecule, arriving from an arbitrary direction bonds to the dendrimer, after encounter another one leaves the dendrimer from an arbitrary part. In this case there is no obstruction at all, *i.e.* $\varphi = 0$ in eqn (11).

The number of water molecules determined in this way is defined as *the average number of water molecules traveling with the rate of dendrimer but exchanging fast with bulk water on the diffusion time scale*. It is not straightforward which model is the closest to the reality, in other words, what is the connection between the determined number of water molecules in this way and the hydration number? The hydration number, by definition, is the total number of water molecules in one dendrimer molecule. Table 3 shows that for E5.NH₂ we obtained values from 2000 to 3700 depending on the model. In the literature the experimental and calculated numbers of water molecules in E5.NH₂ come also from different approximations. Atomistic MD calculations report 2820 (371 buried, 1002 surface and 1448 bulk) water molecules in one E5.NH₂ at high pH based on the calculated solvent excluded surface and well-defined spatial domains for water molecules illustrated in Scheme 2.^{13,15} This model is very similar to our second model (non-penetrable skeleton), and the number of water molecules obtained is similar.

Li *et al.* calculate hydration numbers by subtracting the self-volume occupied by the constituent component (V_M) from total volumes determined by SANS experiments, using different boundary radii. When they use R_{HS} from the “fuzzy ball” model as 2.5 nm the number of water molecules is 1024, while in the case of $R_{HS} = 3$ nm it is about 1700.⁵⁸ These numbers agree well

with the MD calculations, if we count only the “buried” and “surface” water molecules which are present in the “inside cavities” of E5.NH₂.¹³

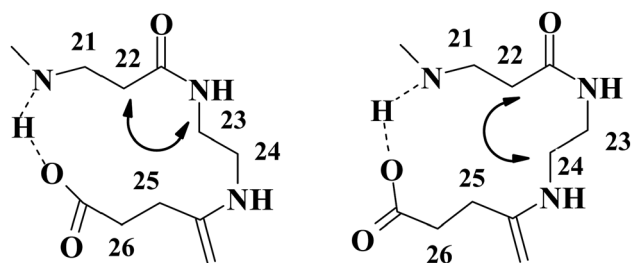
We also made some simple calculations in order to judge the reality of our results. The volume of one water molecule is about 30 Å³ and the total volume of 3700 water molecules is approximately 110 nm³.⁸ The volume of the organic skeleton from constituent atoms is 37 nm³ for E5.NH₂ therefore the hydrated volume is 147 nm³.⁵⁹ The measured hydrodynamic volume from the diffusion coefficient is 120 nm³ that is we over-estimate the volume by 27 nm³ with the third model. This extra volume corresponds to a ~ 0.2 nm extra radius beyond R_H . Since the radius of one water molecule is 0.2 nm it means that there is about one half “molecular layer” of water, about 900 molecules, around an E5.NH₂, which does not belong to the hydrodynamic volume of E5.NH₂. A possible explanation is the hydrogen bonding capability of 117 primary amines. They can bond to almost 400 water molecules and can order more on the “surface” of the fully hydrated dendrimer that is decrease the average value of D_w^{obs} .⁷³ However if we use the second model we have to count only 3100 water molecules (Table 3) therefore we do not have to suppose this “water layer” around E5.NH₂ since the calculated volume is 130 nm³, which is in better agreement with the volume from D_d^0 . The difference (10 nm³) corresponds to about 300 molecules, which may be attributed to the several approximations in the calculations.

The hydrodynamic volume of E5.SAH is 160 nm³ from the diffusion experiment. The organic skeleton is 53 nm³ and completing with the volume of 3600 water molecules according to the third “no obstruction” model the total volume is 161 nm³ in temptingly good agreement with the experimental one. Therefore E5.SAH does not seem to have the “outer” water layer. With the second model the hydrodynamic volume is 137 nm³, which is about 20 nm³ less than what we measured. It means that there should be water molecules in E5.SAH, approximately 700, which contribute to the hydrodynamic volume but do not exchange with the bulk. The explanation may be the following. At the self-pH of E5.SAH the tertiary nitrogens are mainly protonated while carboxylate groups are deprotonated. In this case the inside hydrogen bonds between these two groups, COO[−]⋯H⁺⋯N(*terc*), are probably superior to the COO[−]⋯HOH interaction. This backfolding results in a more crowded molecule than E5.NH₂ which may entrap some water molecules and slows down their exchange with the bulk. The 2D NOESY NMR spectrum indicates the existence of this inside H-bond interaction for E5.SAH in Fig. 7. According to the structure shown in Scheme 3 the formation of the ring may bring closer 24 or 23 to 22 than to 25. The rectangular frame drawn within the 2D sheet shows the strong crosspeak between protons 22 and 23 or 24 (Scheme 1).

Backfolding in Scheme 3 may also be the reason that the number of water molecules, calculated with the third model, in E5.SAH is not larger than that in E5.NH₂ although the former has a larger hydrodynamic radius. The difference in the hydrodynamic radius of E5.NH₂ and E5.SAH is only 320 pm. It is much less than 5–600 pm, the length of succinic acid from a simple geometric assumption. This is also an indication of backfolding.



Fig. 7 2D NOESY spectrum of the PAMAM_E5.SAH dendrimer with a mixing time of 200 ms in D₂O at self-pH. The numbers on the spectrum correspond to Schemes 1 and 3.



Scheme 3 Possible “back folding” orientations which lead to nOe crosspeaks between protons 22 and 23 (left) and 22 and 24 (right) indicated by arrows.

The second model reproduces the experimental number of water molecules in the E5.NH₂ molecule but underestimates the same in E5.SAH. The organic skeleton of dendrimers obviously excludes water molecules from certain volumes of water therefore its obstruction effect cannot be neglected. However we have difficulties to explain why this model is not valid for E5.SAH. The third “no obstruction” model fits to E5.SAH and it is easier to explain why it overestimates the number of water molecules in E5.NH₂. According to our knowledge this type of approximation has not been suggested for the behavior of hydration of soft colloids. However we note that the theoretical description of the interaction between soft or ultra-soft colloids with water is not as elaborated as the description of hard colloids. The reality of this model is under investigation in our lab by using further experiments.

No pH dependence of water diffusion was found in these experiments (Fig. 5) in spite of theoretical predictions of the change of hydration with pH.¹³ However as we indicated above changes in the ionic medium could mask the pH dependence of the water diffusion by changing the bulk viscosity. The complete understanding of structure, size and hydration as a function of pH is still a challenging task and experiments are underway in our laboratory.

Conclusions

We report here the experimental concentration dependence of the apparent self-diffusion coefficient (D_a) of the PAMAM_E5.NH₂ dendrimer and its succinamic derivative (PAMAM_E5.SAH) as determined by PGSE NMR in dilute solutions (0.1–2 mM). From the extrapolation of diffusion coefficient to zero concentration the hydrodynamic radii, R_H , of E5.NH₂ and E5.SAH were found to be 3.05 nm and 3.37 nm respectively. From the slope of the volume fraction dependence of the apparent diffusion coefficient we concluded that generation 5 hydrated PAMAM dendrimers behave like hard sphere colloids with an “effective hard sphere” radius equal to their R_H in dilute aqueous solutions.

Approximately 3600–3700 water molecules are under the influence of one dendrimer molecule if no obstruction effect is considered. This number is higher than that predicted from MD calculations or deduced from SANS experiments for E5.NH₂. We explain the differences as follows: MD calculations consider only water molecules that fulfill narrowly predefined conditions, *i.e.* the pre-given distance from the “surface” of the dendrimer, positions inside or outside of the dendrimer; from SANS experiments those water molecules can only be calculated that occupy the “inside cavities” of the dendrimer molecules, therefore SANS does not deal with the “slow” water molecules around E5.NH₂. However, the PGSE NMR method can detect water molecules bound weakly to terminal NH₂ groups of PAMAM_E5.NH₂. In the case of PAMAM_E5.SAH the number of water molecules determined corresponds exactly to the difference between the hydrodynamic volume and the self-volume of the dendrimer if we use the “no obstruction” model. However if we take into account the obstruction of the organic skeleton we had to suggest the existence of non-exchanging water molecules of structural origin. Both models have advantages but the theoretical basis of the “no obstruction” model is weaker. From these results we can conclude that PGSE NMR, supported by potentiometry and high resolution NMR, appears to be well suited to study and describe the hydration and dynamic behavior of dissolved dendritic macromolecules in water and opens a new route for understanding the hydration dynamics of soft colloids.

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